A Frobenius-cyclic quartic transfer at GF(8)

Audited supplement | 26 October 2026 | exact source, subgroup map, route bounds, and one-target cost

Claim at a glance

An exact plane quartic outside the known direct cubic-cover order closure admits three elliptic degree-two quotients over GF(512). All nine normalized ordinary source curves in its order-554 isogeny class have evaluated degree-two isogeny paths to one quotient. A traced divisor pullback preserves the order-277 subgroup. The frozen relation attack recovers the exact source scalar, but measured total time is far above rho. This is a new exact witness and evaluated route in this project, not a newly classified geometric family or a demonstrated vulnerability.

Exact curve and fields

Let k=GF(8) with a^3+a+1=0, K=GF(512) with w^9+w^4+1=0, and a=w^73. The smooth plane quartic C/k is:

```
x^4 + a^2 x^3y + (a^2+a)x^2y^2 + axy^3 + (a+1)y^4
+ a^2 x^3z + (a^2+a+1)x^2yz + (a^2+1)y^3z
+ (a^2+1)xyz^2 + (a^2+a)y^2z^2 + (a^2+a+1)xz^3
+ a^2 yz^3 + (a^2+a+1)z^4 = 0.
```

Independent point counts are #C(k)=9, #C(GF(64))=65, and #C(K)=636. Its Weil polynomial is T^6+41T^3+512, so #Jac(C)(k)=554=2*277. Magma verifies smoothness, the counts, and geometric automorphism group C2 x C2; the three involutions are defined over K.

Explicit quotient and source

Set X=x, Y=w^132 x+w^68 y+w^179 z, Z=x+z, U=X^2+XY. The regular degree-two map pi:C_K to D sends [x:y:z] to [Y^2:YZ:Z^2:U]. In coordinates [t0:t1:t2:t3], D is the smooth genus-one intersection:

```
t1^2+t0*t2=0,
t3^2+(w^161*t0+w^297*t1+w^83*t2)*t3
 +w^458*t0^2+w^234*t0*t1+w^366*t1^2
 +w^498*t1*t2+w^310*t2^2=0.
```

Magma verifies #D(K)=554 and identifies its elliptic model with E_105: v^2+uv=u^3+u^2+b, b=105 in Conway integer encoding. The explicit isomorphism to the Magma quotient elliptic coordinates is u'=w^418u+w^424, v'=w^116v+w^135u+w^167.

Subgroup-preserving transfer

The nine source b values are 14, 84, 105, 222, 235, 408, 437, 454, and 493. Checked degree-two isogenies follow the cycle 14 -> 493 -> 222 -> 235 -> 105 -> 454 -> 437 -> 408 -> 84 -> 14. The longest selected route into 105 has eight edges and degree at most 256, coprime to 277.

For each source E_b, let u_b be its checked isogeny path into E_105 followed by the quotient isomorphism. The map on K-rational points is

```
Phi_b(P) = Tr_(K/k)(pi^*(u_b(P))) in Jac(C)(k).
```

Pullback followed by norm is multiplication by 2, so pullback is injective on odd-order torsion. The three elliptic factors give a product-to-Jacobian isogeny with 2-primary kernel, and k-Frobenius cycles the three factors. Thus the trace of a nonzero order-277 point remains nonzero. The exact finite-field replay checks this statement on the chosen source pair and all nine isogeny paths.

Exact source pair and checked fibers

On E_105, G=(w^478,w^433), Q=(w^77,w^119)=[7]G, and ord(G)=277. The fibers of the quotient map used for the Sage divisor replay are:

```
pi^-1(O) = {(w^54:w^2:1), (w^75:w^281:1)}
pi^-1(G) = {(w^164:w^2:1), (w^262:w^140:1)}
pi^-1(Q) = {(w^328:w^211:1), (w^109:1:0)}
```

The replay forms both degree-zero pullback classes, sums their three Frobenius conjugates, checks nonzero 277-torsion, and verifies Phi_105(Q)=7 Phi_105(G). Lean checks finite arithmetic plus abstract push-pull and injective-product nonzero lemmas. It does not formalize the geometric or finite-field computations.

General structural criterion

If a smooth nonhyperelliptic genus-three curve over k=GF(q), characteristic 2, has geometric C2 x C2 involutions forming a Frobenius 3-cycle, its elliptic quotients are defined over K=GF(q^3). If one quotient has K-trace t, then:

```
P_C(T)=T^6-t*T^3+q^3;  #Jac(C)(k)=q^3+1-t
#C(k)=q+1;  #C(GF(q^2))=q^2+1
#C(K)=q^3+1-3t;  |t| <= 2*q^(3/2).
```

For odd r dividing q^3+1-t, a source with an evaluated isogeny of degree prime to r into a quotient obtains an injective order-r transfer. These are eligibility and point-count bounds, not a bound proving that a discrete-log attack beats rho. Here q=8, t=-41, r=277.

Finite census and measured cost

Among 1,022 normalized ordinary GF(512) source models, 324 lie in the known direct-B cubic-cover order closure. Nine further models of order 554 have the explicit quartic route above. Their union has 333 models; 689 are outside these two evaluated route closures. This is a bound on the named routes, not a universal bound on all covers or on all vulnerable curves.

The panel also checks exact quartics at source orders 482, 524, 548, and 556. Each has trivial geometric automorphism group, excluding a separable degree-two elliptic quotient for that exact quartic. It does not exclude another polarization or a higher-degree map in its isogeny class.

Frozen same-target benchmark

Stage	Seconds
Curve and Jacobian setup	7.709412
Transfer of both targets	14.447603
Factor-base construction	31.386549
Relations and target decompositions	291.752011
Linear algebra	0.000556
Source verification	0.034788
Full one-target attack	345.331173
Rho median, 32 successful trials	0.000554

The deterministic relation walk used 400 Jacobian additions and retained 189 collision relations. Both transferred targets were decomposed; the relation matrix had a one-dimensional right kernel. Independent audit recomputes the kernel constraints and the scalar 7 from the two decompositions. The source check confirms 7G=Q. The full Sage implementation took about 623,000 times the same-target rho median in wall time. This ratio is implementation-specific and is not a larger-parameter complexity claim.

Prior-work boundary

Nart and Ritzenthaler already classify characteristic-two smooth plane quartics and their Klein-four elliptic quotients. Menezes and Teske already discuss non-subfield source classes of order 0 or 2 modulo 8 under an isogeny-walk assumption; 554 is in that arithmetic set. The exact order-554 quartic is absent from its current LMFDB class listing, but bounded equation searches cannot establish publication novelty. Distinguish (1) new exact equation, (2) new eligible family, and (3) new attack method: this result supports the first label and a newly evaluated route within this project, not the second or third.

Primary references and evidence

Nart-Ritzenthaler (2004); Nart-Ritzenthaler (2009); Menezes-Teske (2004); LMFDB order-554 class. Full scripts, public Magma receipts, relation rows, Lean source, and hash-linked audit are in the companion research directory and PR #2651.